

Tiffany Bao

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INFORMATION Arizona State University
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PROFESSIONAL **Associate Professor, Arizona State University** 2025 August – present
APPOINTMENTS **Assistant Professor, Arizona State University** 2018 – present

EDUCATION **Doctor of Philosophy, Carnegie Mellon University** 2012 – 2018

PUBLICATIONS Teamwork Makes the Defense Work: Comprehensive Vulnerability Defense Resource Allocation.
Siyu Liu, Rida Bazzi, Fei Fang, Tiffany Bao.
In Proceedings of the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS '25).

“It’s almost like Frankenstein”: Investigating the Complexities of Scientific Collaboration and Privilege Management within Research Computing Infrastructures.

Souradip Nath, Ananta Soneji, Jaejong Baek, Tiffany Bao, Adam Doupé, Carlos Rubio-Medrano, Gail-Joon Ahn.

In Proceedings of the 46th IEEE Symposium on Security and Privacy (Oakland '25).

ScamNet: Toward Explainable Large Language Model-Based Fraudulent Shopping Website Detection.

Marzieh Bitaab, Alireza Karimi, Zhuoer Lyu, Ahmadreza Mosallanezhad, Adam Oest, Ruoyu Wang, Tiffany Bao, Yan Shoshitaishvili, Adam Doupé.

In Proceedings of the AAAI Conference on Artificial Intelligence (AAAI '25).

System Register Hijacking: Compromising Kernel Integrity By Turning System Registers Against the System.

Jennifer Miller, Manas Ghandat, Kyle Zeng, Hongkai Chen, Abdelouahab Habs Benchikh, Tiffany Bao, Ruoyu Wang, Adam Doupé, Yan Shoshitaishvili.

In Proceedings of the 34th USENIX Security Symposium (USENIX '25).

SCAMMAGNIFIER: Piercing the Veil of Fraudulent Shopping Website Campaigns.

Marzieh Bitaab, Alireza Karimi, Zhuoer Lyu, Adam Oest, Dhruv Kuchhal, Muhammad Saad, Gail-Joon Ahn, Ruoyu Wang, Tiffany Bao, Yan Shoshitaishvili, Adam Doupé.

In *Proceedings of the Network and Distributed System Security Symposium (NDSS '25)*.

Fuzz to the Future: Uncovering Occluded Future Vulnerabilities via Robust Fuzzing.

Arvind S Raj, Wil Gibbs, Fangzhou Dong, Jayakrishna Menon Vadayath, Michael Tompkins, Steven Wirsz, Yibo Liu, Zhenghao Hu, Chang Zhu, Gokulkrishna Praveen Menon, Brendan Dolan-Gavitt, Adam Doupé, Ruoyu Wang, Yan Shoshitaishvili, Tiffany Bao.

In *Proceedings of the 2024 ACM SIGSAC Conference on Computer and Communications Security (CCS '24)*.

TYGR: Type Inference on Stripped Binaries using Graph Neural Networks.

Chang Zhu, Ziyang Li, Anton Xue, Ati Priya Bajaj, Wil Gibbs, Yibo Liu, Rajeev Alur, Tiffany Bao, Hanjun Dai, Adam Doupé, Mayur Naik, Yan Shoshitaishvili, Ruoyu Wang, Aravind Machiry.

In *Proceedings of the 33rd USENIX Security Symposium (USENIX '24)*.

Take a Step Further: Understanding Page Spray in Linux Kernel Exploitation.

Ziyi Guo, Dang K Le, Zhenpeng Lin, Kyle Zeng, Ruoyu Wang, Tiffany Bao, Yan Shoshitaishvili, Adam Doupé, Xinyu Xing.

In *Proceedings of the 33rd USENIX Security Symposium (USENIX '24)*.

Ahoy SAILR! There is No Need to DREAM of C: A Compiler-Aware Structuring Algorithm for Binary Decompilation.

Zion Leonahenahe Basque, Ati Priya Bajaj, Wil Gibbs, Jude O’Kain, Derron Miao, Tiffany Bao, Adam Doupé, Yan Shoshitaishvili, Ruoyu Wang.

In *Proceedings of the 33rd USENIX Security Symposium (USENIX '24)*.

AirTaint: Making Dynamic Taint Analysis Faster and Easier.

Qian Sang, Yanhao Wang, Yuwei Liu, Xiangkun Jia, Tiffany Bao, Purui Su.

In *Proceedings of the IEEE Symposium on Security and Privacy (Oakland '24)*.

“Watching over the shoulder of a professional”: Why hackers make mistakes and how they fix them.

Irina Ford, Ananta Soneji, Faris Bugra Kokulu, Jayakrishna Vadayath, Zion Leonahenahe Basque, Gaurav Vipat, Adam Doupé, Ruoyu Wang, Gail-Joon Ahn, Tiffany Bao, Yan Shoshitaishvili.

In *Proceedings of the IEEE Symposium on Security and Privacy (Oakland '24)*.

RetSpill: Igniting User-Controlled Data to Burn Away Linux Kernel Protections.

Kyle Zeng, Zhenpeng Lin, Kangjie Lu, Xinyu Xing, Ruoyu Wang, Adam Doupé, Yan Shoshitaishvili, Tiffany Bao.

In *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security (CCS '23)*.

Greenhouse: Single-Service Rehosting of Linux-Based Firmware Binaries in User-Space Emulation.

Hui Jun Tay, Kyle Zeng, Jayakrishna Menon Vadayath, Arvind S Raj, Audrey Dutcher, Tejesh Reddy, Wil Gibbs, Zion Leonahenahe Basque, Fangzhou Dong, Zack Smith, Adam Doupé, Tiffany Bao, Yan Shoshitaishvili, Ruoyu Wang.

In *the 32nd USENIX Security Symposium (USENIX '23)*.

BEYOND PHISH: Toward Detecting Fraudulent e-Commerce Websites at Scale. Marzieh Bitaab, Haehyun Cho, Adam Oest, Zhuoer Lyu, Wei Wang, Jorij Abraham, Ruoyu Wang, Tiffany Bao, Yan Shoshitaishvili, Adam Doupé.

In *Proceedings of the IEEE Symposium on Security and Privacy (Oakland '23)*.

Toss a Fault to Your Witcher: Applying Grey-box Coverage-guided Mutational Fuzzing to Detect SQL and Command Injection Vulnerabilities.

Erik Trickel, Fabio Pagani, Chang Zhu, Lukas Dresel, Giovanni Vigna, Christopher Kruegel, Ruoyu Wang, Tiffany Bao, Yan Shoshitaishvili, Adam Doupé.

In *Proceedings of the IEEE Symposium on Security and Privacy (Oakland '23)*.

Cyber Deception: Techniques, Strategies, and Human Aspects.

Tiffany Bao, Milind Tambe, Cliff Wang.

Springer Nature.

Mitigating Threats Emerging from the Interaction between SDN Apps and SDN (Configuration) Datastore.

Sana Habib, Tiffany Bao, Yan Shoshitaishvili, Adam Doupé.

Proceedings of the 2022 on Cloud Computing Security Workshop.

Playing for K(H)eaps: Understanding and Improving Linux Kernel Exploit Re-

liability.

Kyle Zeng, Yueqi Chen, Haehyun Cho, Xinyu Xing, Adam Doupé, Yan Shoshitaishvili, Tiffany Bao.

In the 31st USENIX Security Symposium (USENIX '22).

Arbiter: Bridging the Static and Dynamic Divide in Vulnerability Discovery on Binary Programs.

Jayakrishna Vadayath, Moritz Eckert, Kyle Zeng, Nicolaas Weideman, Gokulkrishna Praveen Menon, Yanick Fratantonio, Davide Balzarotti, Adam Doupé, Tiffany Bao, Ruoyu Wang, Christophe Hauser, Yan Shoshitaishvili.

In the 31st USENIX Security Symposium (USENIX '22).

Expected Exploitability: Predicting the Development of Functional Vulnerability Exploits.

Octavian Suci, Connor Nelson, Zhuoer Lyu, and Tiffany Bao, Tudor Dumitras.
In the 31st USENIX Security Symposium (USENIX '22).

"Flawed, but like democracy we don't have a better system": The Experts' Insights on the Peer Review Process of Evaluating Security Papers.

Ananta Soneji, Faris Bugra Kokulu, Carlos Rubio-Medrano, Tiffany Bao, Ruoyu Wang, Yan Shoshitaishvili, Adam Doupé.

In Proceedings of the IEEE Symposium on Security and Privacy (Oakland '22).

ViK: practical mitigation of temporal memory safety violations through object ID inspection.

Haehyun Cho, Jinbum Park, Adam Oest, Tiffany Bao, Ruoyu Wang, Yan Shoshitaishvili, Adam Doupé, Gail-Joon Ahn.

Proceedings of the 27th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS '22).

Favocado: Fuzzing Binding Code of JavaScript Engines Using Semantically Correct Test Cases.

Sung Ta Dinh, Haehyun Cho, Kyle Martin, Adam Oest, Yihui Zeng, Alexandros Kapravelos, Tiffany Bao, Ruoyu Wang, Yan Shoshitaishvili, Adam Doupé, Gail-Joon Ahn.

In Network and Distributed System Security Symposium (NDSS '21).

Having Your Cake and Eating It: An Analysis of Concession-Abuse-as-a-Service
Zhibo Sun, Adam Oest, Penghui Zhang, Carlos Rubio-Medrano, Tiffany Bao, Ruoyu Wang, Ziming Zhao, Yan Shoshitaishvili, Adam Doupé, Gail-Joon Ahn.

In *the 30th USENIX Security Symposium (USENIX '21)*.

CrawlPhish: Large-scale Analysis of Client-side Cloaking Techniques in Phishing.

Penghui Zhang, Adam Oest, Haehyun Cho, Zhibo Sun, RC Johnson, Brad Wardman, Shaown Sarker, Alexandros Kapravelos, Tiffany Bao, Ruoyu Wang, Yan Shoshitaishvili, Adam Doupé, Gail-Joon Ahn.

In *Proceedings of the IEEE Symposium on Security and Privacy (Oakland '21)*.

SyML: Guiding Symbolic Execution Toward Vulnerable States Through Pattern Learning.

Nicola Ruaro, Kyle Zeng, Lukas Dresel, Mario Polino, Tiffany Bao, Andrea Continella, Stefano Zanero, Christopher Kruegel, Giovanni Vigna.

In *the 24th International Symposium on Research in Attacks, Intrusions and Defenses (RAID '21)*.

SoK: Everything You Ever Wanted to Know About Bitcoin Mixers (But Were Afraid to Ask)

Jaswant Pakki, Yan Shoshitaishvili, Ruoyu Wang, Tiffany Bao, Adam Doupé.

In *Financial Cryptography and Data Security (FC '21)*.

MuTent: Dynamic Android Intent Protection with Ownership-Based Key Distribution and Security Contracts.

Pradeep Kumar Duraisamy Soundrapandian, Jaejong Baek, Tiffany Bao, Yan Shoshitaishvili, Adam Doupé, Ruoyu Wang, Gail-Joon Ahn.

In *Hawaii International Conference on System Sciences (HICCS '21)*.

Best Minitrack Paper Award.

HoneyPLC: A Next-Generation Honeypot for Industrial Control Systems

Efren López-Morale, Carlos Rubio-Medrano, Adam Doupé, Yan Shoshitaishvili, Ruoyu Wang, Tiffany Bao, Gail-Joon Ahn.

In *Proceedings of the 2020 ACM SIGSAC Conference on Computer and Communications Security (CCS '20)*.

Exploiting Uses of Uninitialized Stack Variables in Linux Kernels to Leak Kernel Pointers.

Haehyun Cho, Jinbum Park, Joonwon Kang, Tiffany Bao, Ruoyu Wang, Yan Shoshitaishvili, Adam Doupé, Gail-Joon Ahn.

In *the 14th USENIX Workshop on Offensive Technologies (WOOT '20)*.

Not All Coverage Measurements Are Equal: Fuzzing by Coverage Accounting for Input Prioritization.

Yanhao Wang, Xiangkun Jia, Yuwei Liu, Tiffany Bao, Dinghao Wu, and Purui Su.

In the *Network and Distributed System Security Symposium (NDSS '20)*.

Scam Pandemic: How Attackers Exploit Public Fear through Phishing.

Marzieh Bitaab, Haehyun Cho, Adam Oest, Penghui Zhang, Zhibo Sun, Rana Pourmohamad, Doowon Kim, Tiffany Bao, Ruoyu Wang, Yan Shoshitaishvili, Adam Doupé, Gail-Joon Ahn. In *Symposium on Electronic Crime Research (ECrime '20)*.

Cyber Autonomy in Software Security: Techniques and Tactics (Book Chapter).

Tiffany Bao and Yan Shoshitaishvili.

In *Game Theory and Machine Learning for Cyber Security*.

Matched and Mismatched SOCs: A Qualitative Study on Security Operations Center Issues.

Faris Bugra Kokulu, Ananta Soneji, Tiffany Bao, Yan Shoshitaishvili, Ziming Zhao, Adam Doupé, and Gail-Joon Ahn.

In *Proceedings of the 2019 ACM SIGSAC Conference on Computer and Communications Security (CCS '19)*.

Understanding and Predicting Private Interactions in Underground Forums.

Zhibo Sun, Carlos E. Rubio-Medrano, Ziming Zhao, Tiffany Bao, Adam Doupé, and Gail-Joon Ahn.

In the *9th ACM Conference on Data and Application Security and Privacy (CODASPY '19)*.

Your Exploit is Mine: Automatic Shellcode Transplant for Remote Exploits.

Tiffany Bao, Ruoyu Wang, Yan Shoshitaishvili, and David Brumley.

In *Proceedings of the 38th IEEE Symposium on Security and Privacy (Oakland '17)*.

How Shall We Play a Game: A Game-theoretical Model for Cyber-warfare.

Tiffany Bao, Yan Shoshitaishvili, Ruoyu Wang, Christopher Kruegel, Giovanni Vigna and David Brumley.

In *Proceedings of the 30th IEEE Computer Security Foundations (CSF '17)*.

National Security Agency's Annual Scientific Paper Award.

Security is a Game.

Tiffany Bao.

In *2017 USENIX Summit on Hot Topics in Security (HotSec '17)*.

Awarded Talk.

A Game-theoretical Model for Cyber-warfare Games. (Invited Poster)

Tiffany Bao, Yan Shoshitaishvili, Ruoyu Wang and David Brumley.

In *the 8th Workshop on Computational Cybersecurity in Compromised Environments (C3E '16)*.

ByteWeight: Learning to Recognize Functions in Binary Code.

Tiffany Bao, Jonathan Burket, Maverick Woo, Rafael Turner, and David Brumley.

In *Proceedings of the 23rd USENIX Security Symposium (USENIX '14)*.

Type-based Dynamic Taint Analysis Technology.

Libo Chen, Jianwei Zhuge, Fan Tian, Tiffany Bao, and Xun Lu. In *Tsinghua Science and Technology Journal*, 2012.

Research of Technology for Type-based Dynamic Taint Analysis.

Libo Chen, Jianwei Zhuge, Fan Tian, Tiffany Bao, and Xun Lu. In *Proceedings of the 5th Conference of Vulnerability Analysis and Risk Assessment (VARA '12)*.

Outstanding Paper Award.

FUNDED
PROJECTS

CAREER: Achieving Autonomous Symbolic Execution through Learning from Humans 2025-2030

National Science Foundation (NSF).

Awarded funding: \$500,000

SENPAI: Strategic Exploration, Navigation, and Patching of Abstracted Integrations for Cyber-Physical Systems 2024-2027 DOD-DARPA: Information Innovation Office (DARPA MTO).

Awarded funding: \$7,960,690

EURYALE: Combating Emergent Execution with a GLANCE 2025-2026 DOD-DARPA: Information Innovation Office (IIO).

Awarded funding: \$1,523,727

Dynamic and Personalized Deception for Proactive Cyber Defense 2024-2027
DOD-USAF-AFRL: Air Force Office of Scientific Research (AFOSR).
Awarded funding: \$599,968

American Cybersecurity Education (ACE) Institute 2024-2026
DOD: Defense Advanced Research Projects Agency (DARPA).
Awarded funding: \$4,462,558

RxCRS: Reliable and eXplainable Cyber Reasoning System for Digital Health Security 2023-2025
HHS-NIH: Advanced Research Projects Agency for Health (ARPA-H).
Awarded funding: \$6,564,290

Assessing Cyber Training by Cyber Operation Mistakes 2023-2026
DOD-NAVY: Office of Naval Research (ONR).
Awarded funding: \$900,135

SaTC: CORE: Medium: Symbolizing Viability: Paving the Road to Practical Symbolic Execution 2023-2027
National Science Foundation (NSF).
Awarded funding: \$1,199,992

NSF-SFS: Arizona Cyber Defense Scholarship 2017-2026
National Science Foundation (NSF).
Awarded funding: \$4,297,774
Cybersecurity Meta-analysis for USAID Beneficiaries. 2021-2022
US Agency for International Development (USAID).
Awarded funding: \$59,655

Predicting and Assessing Attacks for Open Source Projects. 2021-2022
DOD-DARPA: Information Innovation Office (DARPA IIO).
Awarded funding: \$487,720

SaTC: CORE: Medium: Self-Adaptive Cyber Risk Management 2020 - 2023
via Machine to Machine Economy supported by blockchain and smart contracts technology.
National Science Foundation (NSF).
Awarded funding: \$750,000

VOLT: A Viscous, Orchestrated Lifting and Translation Framework 2020-2023
DOD-DARPA: Information Innovation Office (DARPA IIO).
Awarded funding: \$5,045,355

Human-Assisted Cyber Reasoning Systems and Oppositional 2019-2023
Human Factors.
DOD: National Security Agency (NSA).
Awarded funding: \$5,598,441

CHECRS: Cognitive Human Enhancements for Cyber Reasoning 2018-2022
Systems.
DOD-DARPA: Information Innovation Office (DARPA IIO).
Awarded funding: \$11,730,556

Realizing Cyber Inception: Towards a Science of Personalized 2017-2023
Deception for Cyber Defense.
DOD-ARMY-ARL: Army Research Office (ARO).
Awarded funding: \$598,149

PROFESSIONAL
ACTIVITIES

Chair Member
Research in Attacks, Intrusions and Defenses. Co-Chair. 2025, 2026
Network and Distributed System Security Symposium Poster Session. 2023,
2024
USENIX Security, Associate chair member. 2024
IEEE Security and Privacy, Vice chair member. 2024

Program Committee Member
Network and Distributed System Security Symposium. 2019 - 2023, 2025
ACM SIGSAC Conference on Computer and Communications Security. 2020,
2022-2024
IEEE Security and Privacy. 2022, 2023, 2024
USENIX Security. 2022, 2024, 2025
International Symposium on Research in Attacks, Intrusions and Defenses. 2020
ACM Workshop on Automotive and Aerial Vehicle Security. 2020
(Affiliated with the ACM Conference on Data and Application Security and Pri-

vacy.)

Network and Distributed System Security Symposium. 2019

Annual Computer Security Applications Conference. 2019

Workshop on Binary Analysis Research. 2018, 2019, 2020

(Affiliated with the Network and Distributed System Security Symposium.)

Journal Editorial Board

The Journal of Systems Research, Security Track.

Journal Reviewer

ACM Transactions on information and System Security.

Communications of the ACM.

IEEE Transactions on Software Engineering.

IEEE Transactions on Knowledge and Data Engineering.

CTF Game Organizer

Member of the DEF CON CTF hosting team. 2019 - 2021

Mentor

Google Summer of Code. 2013

SELECTED AWARDS

IEEE Security and Privacy Best Student Paper Award. 2020

National Security Agency's Annual Scientific Paper Award. 2018

Awarded Talk at 2017 USENIX Summit on Hot Topics in Security. 2017

Carnegie Mellon University Presidential Fellowship. 2017